

EARLY GAME INDICATORS

IN PROFESSIONAL LEAGUE OF LEGENDS

Course: COMP-248 Project in Data Science | **Author:** George Andreou | **University of Nicosia**

YouTube Video: <https://www.youtube.com/watch?v=X-VCSw-fySo>

GitHub Repository: <https://github.com/Geoand02/lol-early-game-analysis>

Project Overview

This project analyses professional League of Legends esports match data to identify which early-game indicators (events within the first 15 minutes) most reliably predict the winner. Core research questions: which stats correlate with winning, how reliably early indicators forecast outcomes, whether their importance changes across seasons, and whether competitive regions show different early-game patterns. Data from Oracle's Elixir covers three seasons (2019, 2022, 2025) across the five main regions (LCK, LPL, LEC, LCS, PCS), totalling **53,882 team-level match rows**.

Data Science Process Followed

Data Collection. Three CSV files from Oracle's Elixir, one per season, with consistent column schemas. Each file mixes team-level and player-level rows; only team rows carry objective data.

Data Modeling. A unified DataFrame was built by reindexing each yearly file to a common 34-column schema (15-min stats, first-objectives, side, patch, year, league, game length), concatenated, filtered to team rows, and restricted to the three target seasons.

Data Preprocessing. Engineered binary indicators (*gold_lead*, *xp_lead*, *cs_lead*, *kill_lead*, *got_inhibitor*), a 0–5 *early_score* combining five binary indicators, and categorical buckets for both gold difference and game length. A regional filter (*df_main*) was used for per-league analysis.

Analysis. EDA in Python (pandas, numpy) inside a Jupyter notebook. Correlation analysis, group-by aggregations, mediator-style tables, and false-positive analyses were combined to build a causal-style narrative of how early advantages translate to wins. Matplotlib for visualisation. Per the professor's guidance during Milestone 1, no machine-learning model was built — focus was on depth of analysis rather than predictive modelling.

Technical Challenges Encountered

1. Mixed-row schema (the hardest bug). Initial analysis only saw 39,000 rows and 2019 was silently missing. Diagnosis: Oracle's Elixir mixes player-level rows (positions: top, jng, mid, bot, sup) with team-level rows; objective columns are *NaN* for player rows. A naive `dropna()` destroyed roughly 80% of all rows, disproportionately killing 2019 (less-populated *elementaldrakes* column). Fix: filter on `position == "team"` first, then targeted `dropna(subset=[...])` on essential columns only. This recovered 15,560 rows of 2019 data.

2. Year-column leakage. CSVs contain tournament games that straddle calendar years (e.g. Worlds in January). 2020, 2023 and 2026 leaked in with small samples. Fix: explicit `df["year"].isin([2019, 2022, 2025])` filter after merging.

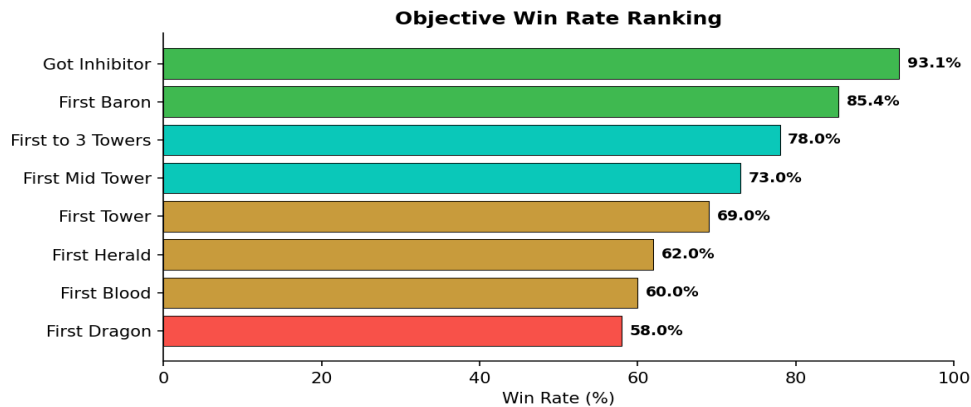
3. Cross-machine path drift. Notebook developed on laptop with hard-coded Windows paths, later moved to a desktop where the file structure differed. A directory walk script located the misplaced data files; absolute paths were updated.

Key Findings

Ten substantive findings emerged from the analysis. The five strongest are summarised below alongside the visualisations that support them.

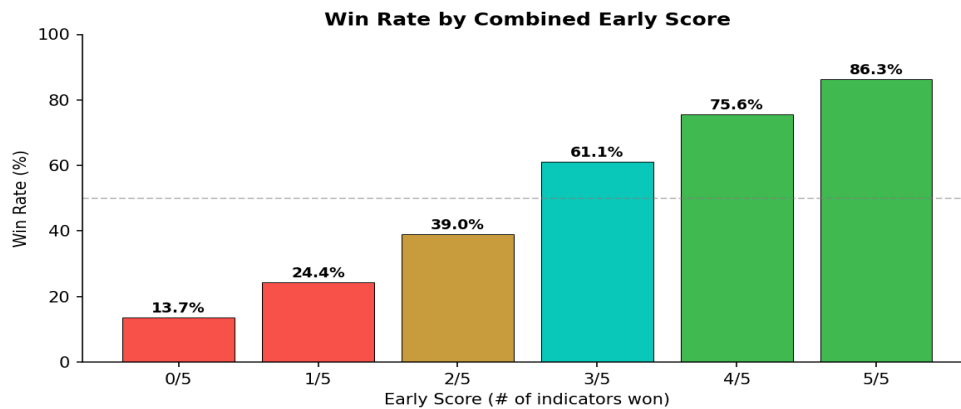
1. Objective Hierarchy

Tower-based objectives consistently outperform kill- and dragon-based ones. First-to-3-towers (78%) is nearly as strong a predictor as first baron (85%), while first blood and first dragon hover around 60% — barely above coin-flip.



2. Combined Early Score Scales Linearly

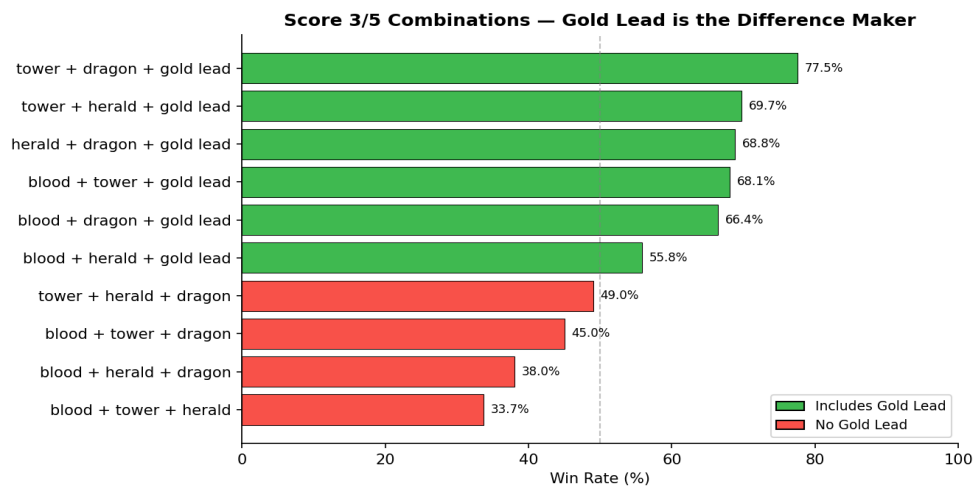
A combined early score (0–5) summing five binary indicators (first blood, first tower, first herald, first dragon, gold lead at 15) reveals near-linear scaling. The 2/5 → 3/5 jump is the threshold that flips a team from underdog to favourite (39% → 61%).



3. Gold Lead is the Universal Premium

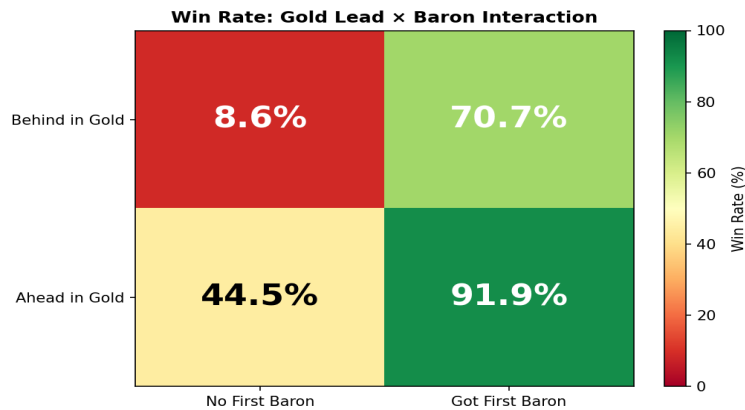
Within every score level, having a gold lead at 15 is worth a remarkably constant **+27 percentage points** of win rate. First blood paired with gold lead adds zero; first dragon adds +15.5pp, first tower +11pp. The premium holds across all 5 main regions (+44 to +59pp) and across all three seasons, confirming it as a universal rule of professional play.

Visualisation of Finding 3 — Same Score, Different Combinations



4. Baron is the Conversion Mechanism

A team ahead in gold at 15 but failing to secure first baron wins only 44.5% — barely above a coin flip. Securing baron raises that to 91.9%. Most strikingly, teams BEHIND in gold who steal baron win 70.7% — an 8.6% → 70.7% swing of **62 percentage points**. Baron is the single most powerful comeback mechanism in pro LoL.



5. All Early Objectives Decay Over Time

Every early-game indicator loses predictive power as games stretch past 35 minutes. Gold lead drops from 97% (sub-25min) to 52% (40+ min). Only inhibitor — a late-game objective — retains decisive power (72% even in 40+ min games). The game effectively "resets" past 35 minutes, explaining why comebacks become viable in long games.

Reflection & Deviations from Proposal

Three changes were made to the original proposal. First, the planned machine-learning classifier was dropped on the professor's advice during Milestone 1, reallocating that time to deeper statistical analysis. Second, late-game champion-composition analysis was abandoned because the selected feature set did not include champion data. Third, KNIME was not used in the final deliverable — once the analysis matured in pandas with matplotlib for visualisation, adding a separate KNIME workflow would have duplicated work without strengthening the findings. All three deviations focused effort on what the data could rigorously support. The biggest unanticipated finding was that gold lead is a near-universal +27pp premium independent of which other objectives are won — a discovery (made while debugging the schema bug) that re-shaped the entire project narrative.